Alpha Diversity

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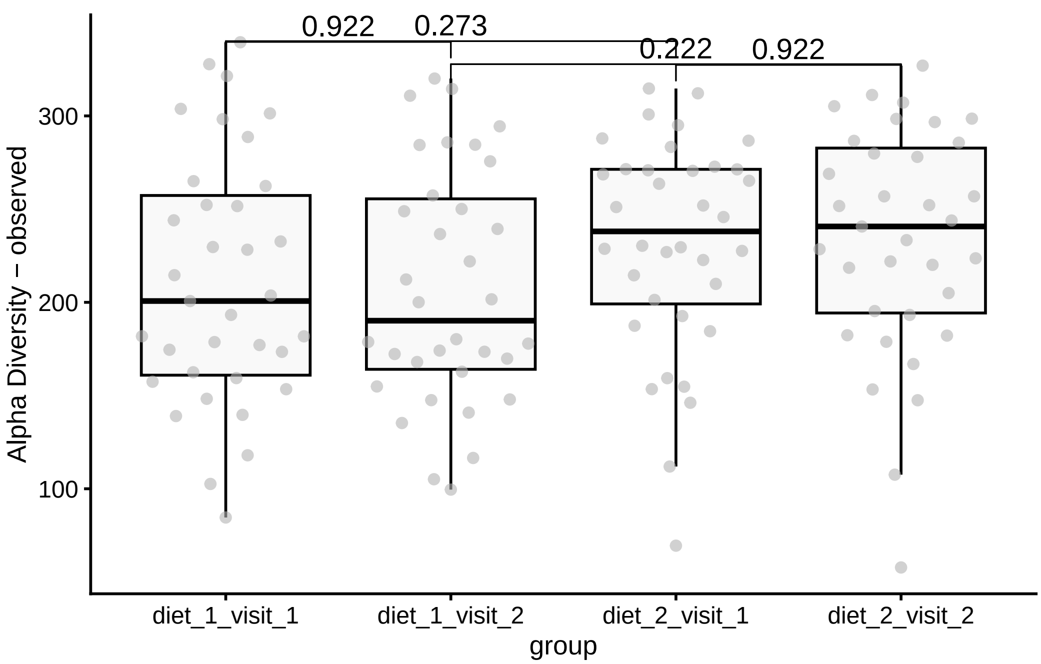
Introduction

This document reports on a microbiota project results of alpha diversity across different diets and visits.

Group-wise comparisons

- **Index:** observed
- **Group variable:** group
- **Multiple testing correction:** fdr

Visualization



	after			before		
Characteristic	**Beta**	**95% CI**[1]	**p-value**	**Beta**	**95% CI**[1]	**p-value**
(Intercept)	144	59, 229	0.001	152	67, 236	<0.001
meal_group			0.3			0.4
1	—	—		—	—	
2	28	-16, 72	0.2	32	-11, 75	0.14
3	30	-14, 73	0.2	27	-15, 70	0.2
4	-3.8	-47, 39	0.9	10	-32, 53	0.6
age	1.2	-0.25, 2.7	0.10	1.0	-0.45, 2.5	0.2

[1]CI = Confidence Interval

Paired test of before and after	**Beta**	**95% CI**[1]	**p-value**
(Intercept)	147	89, 206	<0.001
meal_group			0.11
1	—	—	
2	28	-2.7, 58	0.074
3	28	-2.5, 58	0.072
4	2.2	-28, 32	0.9
age	1.2	0.13, 2.2	0.026

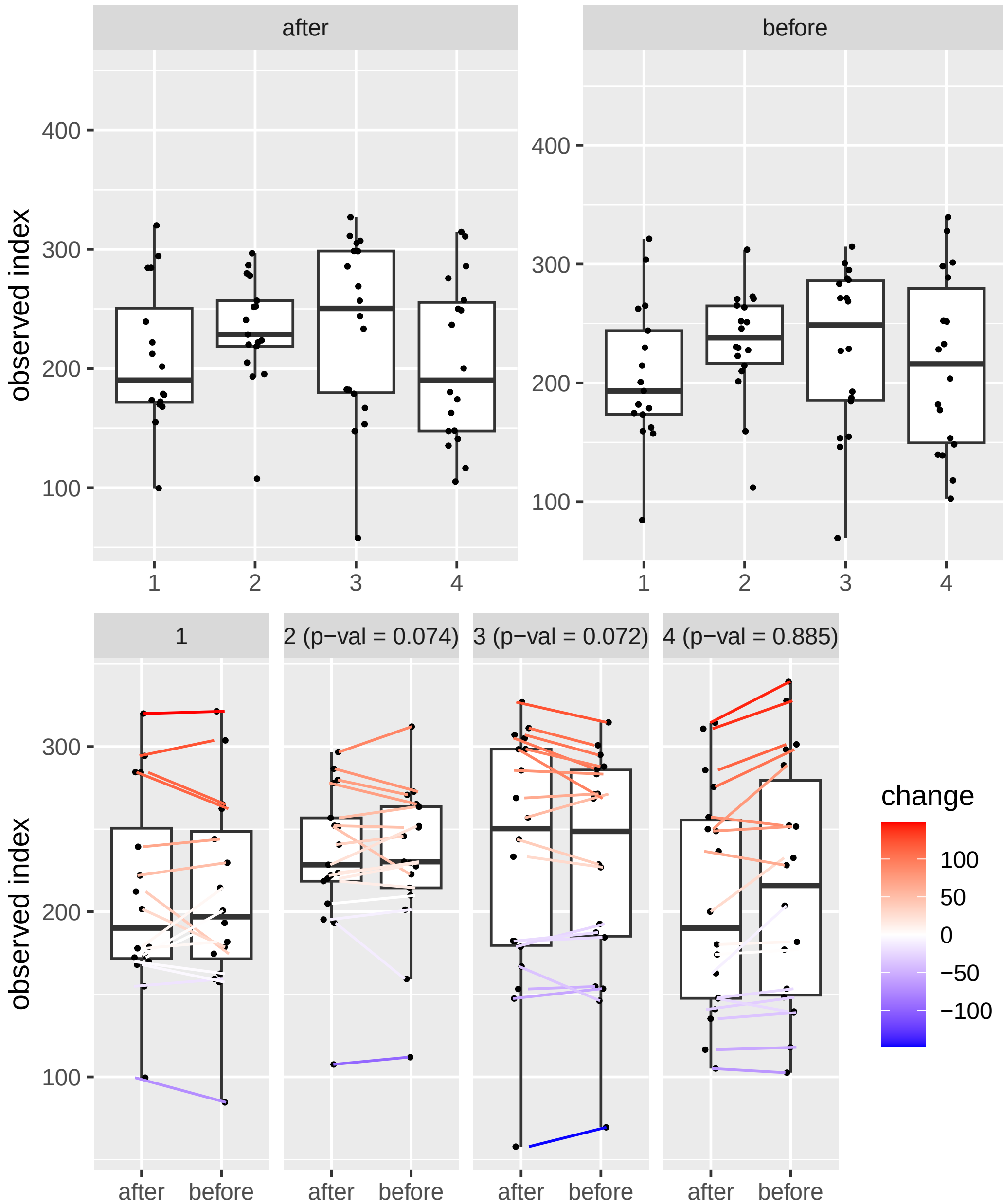
[1]CI = Confidence Interval

Table of the top findings

Table 1: Table of alpha diversity comparisons

group1	group2	mean1	mean2	logFC	p.adj
diet_1_visit_1	diet_1_visit_2	211.16	207.15	-0.03	0.922
diet_2_visit_1	diet_2_visit_2	231.52	233.18	0.01	0.922
diet_1_visit_1	diet_2_visit_1	211.16	231.52	0.13	0.273
diet_1_visit_2	diet_2_visit_2	207.15	233.18	0.17	0.222

Alternate analyses



Data Summary

```
class: TreeSummarizedExperiment
dim: 1700 140
```

```
metadata(0):
assays(2): counts relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(13): sample id ... shannon group
reducedDimNames(0):
mainExpName: NULL
altExpNames(10): PrevalentGenus PrevalentSpecies ... species strain
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL
```

Version info

```
R version 4.4.2 (2024-10-31)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 24.04.1 LTS

Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0

locale:
 [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
 [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
 [9] LC_ADDRESS=C               LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

time zone: Europe/Helsinki
tzcode source: system (glibc)

attached base packages:
[1] stats4    grid      stats     graphics  grDevices utils     datasets
[8] methods   base

other attached packages:
 [1] kableExtra_1.4.0                vegan_2.6-8
 [3] lattice_0.22-6                  permute_0.9-7
```

```
 [5] lubridate_1.9.3                   forcats_1.0.0
 [7] stringr_1.5.1                     purrr_1.0.2
 [9] readr_2.1.5                       tibble_3.2.1
[11] tidyverse_2.0.0                   tidyr_1.3.1
[13] sechm_1.13.1                      ComplexHeatmap_2.21.1
[15] scater_1.33.4                     scuttle_1.15.4
[17] quarto_1.4.4                      patchwork_1.3.0
[19] parameters_0.24.0                 multtest_2.61.0
[21] miaTime_0.1.21                    miaViz_1.15.4
[23] ggraph_2.2.1                      mia_1.15.6
[25] TreeSummarizedExperiment_2.13.0 Biostrings_2.73.2
[27] XVector_0.45.0                    SingleCellExperiment_1.27.2
[29] MultiAssayExperiment_1.31.5       SummarizedExperiment_1.35.3
[31] Biobase_2.65.1                    GenomicRanges_1.57.1
[33] GenomeInfoDb_1.41.2               IRanges_2.39.2
[35] S4Vectors_0.43.2                  BiocGenerics_0.51.3
[37] MatrixGenerics_1.17.0             matrixStats_1.4.1
[39] gridExtra_2.3                     gtsummary_2.0.4
[41] ggsignif_0.6.4                    ggpubr_0.6.0
[43] ggplot2_3.5.1                     DT_0.33
[45] dplyr_1.1.4                       cardx_0.2.2
[47] ANCOMBC_2.8.0

loaded via a namespace (and not attached):
  [1] randomcoloR_1.1.0.1         gld_2.6.6
  [3] nnet_7.3-19                 TH.data_1.1-2
  [5] vctrs_0.6.5                 energy_1.7-12
  [7] digest_0.6.37               png_0.1-8
  [9] shape_1.4.6.1               proxy_0.4-27
 [11] Exact_3.3                   registry_0.5-1
 [13] rbiom_1.0.3                 ggrepel_0.9.6
 [15] bayestestR_0.15.0           parallelly_1.38.0
 [17] mediation_4.5.0             MASS_7.3-61
 [19] reshape2_1.4.4              foreach_1.5.2
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 [23] ggfun_0.1.6                 survival_3.7-0
 [25] doRNG_1.8.6                 commonmark_1.9.1
 [27] memoise_2.0.1               ggbeeswarm_0.7.2
 [29] emmeans_1.10.4              systemfonts_1.1.0
 [31] gmp_0.7-5                   tidytree_0.4.6
 [33] zoo_1.8-12                  GlobalOptions_0.1.2
 [35] gtools_3.9.5                V8_5.0.1
 [37] Formula_1.2-5               datawizard_0.13.0
```

```
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 [93] multcomp_1.4-26             BiocSingular_1.21.4
 [95] slam_0.1-54                 e1071_1.7-16
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```